# Aditya Mitra

Cybersecurity Researcher · Security Engineer · FIDO2/Passkeys Specialist

adityamitra5102@gmail.com · LinkedIn · Website · Google Scholar · ORCID · Credly


I make secure systems. I make systems secure. My work spans passwordless authentication, applied cryptography, and systems security — from the first physical FIDO2 security key with post-quantum cryptography to enterprise authentication infrastructure and open-source security tooling. Erasmus Mundus Scholar in Applied Cybersecurity, active researcher with 15+ publications across IEEE, Springer, and Elsevier, and currently contributing to Nitrokey GmbH's security stack.

## Education

**M.Sc. in Cybersecurity** · Erasmus Mundus Joint Masters (CyberMACS) · 2025 – Present
Kadir Has University, Istanbul, Turkey
*Erasmus Mundus Scholar*

**B.Tech. in Computer Science and Engineering** (Specialization: Robotics) · CGPA: 8.32 · 2020 – 2024
VIT-AP University, Andhra Pradesh, India

## Experience

**Software Development Intern** · Nitrokey GmbH · June 2026 – Present

- Development of Nitrokey 3 management software.

**CTO** · Digital Fortress Pvt. Ltd. · January 2023 – Present

- Architected and built a passwordless authentication backend from scratch in under 3 days, deployed at enterprise scale with SSO and CSP capabilities.
- Led a team of 7 engineers building next-generation FIDO2-based authentication systems.
- Deployed and maintained cloud infrastructure across Azure and Linux-based on-premises servers.

**Researcher** · Center of Excellence in AI and Robotics, VIT-AP · February 2022 – Present

- Researching passwordless authentication, FIDO2/passkeys, post-quantum cryptography, and IoT security.
- 15+ publications across IEEE, Springer, Elsevier, and arXiv.

**Pentester Intern** · Fresh Digital · March 2022 – April 2022

- Conducted penetration testing of production environments; identified compliance gaps and delivered remediation strategies.

## Research Interests

FIDO2 / Passkeys · Post-Quantum Cryptography · Passwordless Authentication · Cryptographic Protocols · Protocol Design · Systems Security · IoT Security · Networking · Decentralized Identity

## Skills

**Languages & Tools:** Python, C, Java, Bash, Git, Linux, CUDA
**Cryptography & Security:** Cryptographic protocol design, FIDO2/WebAuthn, Post-Quantum Cryptography, PKI, ISO 7816
**Platforms & Infra:** Linux, bare-metal deployment, on-premises infrastructure, networking, Azure, SQL
**Libraries:** Flask, python-fido2, pycryptodome, cbor2

## Notable Achievements

- **Vulnerability 1721, FIDO Alliance** — Reported NFC-based MITM attack affecting all NFC FIDO authenticators
- **FIDO Developer Challenge India 2022, Top 3** — FIDO Alliance organized global competition for developers building on FIDO2 specifications
- **Peer Reviewer** — IEEE Access (2 papers), IEEE ICECET 2024, Elsevier Cyber Security and Applications, Springer Telecommunication Systems (2 papers), Springer Nature Computer Science
- **Hackathon Mentor** — Status Code 1 (IIIT Kalyani × IISER Kolkata), FrostHacks 1.0 (Academy of Technology), HexaFalls 1.0 (JIS University)
- **Vice President**, Null Chapter VIT-AP (2022–2023) — organized CTF events, security workshops, FIDO2 demos; awarded Best Member 2021–22

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## Selected Projects

**Mariana's Qubit** *(Ongoing)* · read more
Open-source, anonymous, decentralized internet stack that redefines the OSI model with privacy and censorship resistance as first principles.

**PQC FIDO2 Security Key** · read more
First known implementation of a fully functional physical FIDO2 security key with post-quantum cryptography, built on Raspberry Pi and tested against Google, Windows Hello, and Apple.

**PQRSTUV – Open Source HSM** · read more
Consumer-grade hardware HSM with post-quantum cryptographic guarantees, custom communication protocols, and IETF-aligned standards documentation.

**Loki** *(Published in IEEE Access)* · read more
FIDO2-compliant physical lock for protecting physical assets using RSA and cloud security — among the first physical locks with seamless high-security FIDO2 access control.

## Hackathons & Competitions

Smart India Hackathon 2022 (Winner) · Microsoft Imagine Cup 2022 (India Runner-up) · INEX 2022 (Silver) · ACM ICPC 2021 (Regionalist) · ION<athon> 1.0 (Top 10) · Hackerearth Hack to Enable 2021 (6th)

## Certifications

Cisco Certified Cybersecurity Associate (CCNA Cybersecurity), Cisco Certified Network Associate (CCNA), CompTIA SecurityX (CASP+), CompTIA CySA+, CompTIA Security+, CompTIA Network+, Microsoft SC-900, Microsoft AZ-900, Oracle Java SE 8 OCA, CLA (C Programmer) · All badges on Credly

## Publications

*15+ total — selected below. Full list on Google Scholar.*

- Mitra, A. et al. **Blind Eye: Motion and Obstacle Detection Leveraging Wi-Fi.** IEEE AVSS 2025, pp. 1–7. 10.1109/AVSS65446.2025.11149826
- Mitra, A. & Ghosh, A. **FIDO2: A Comprehensive Study on Passwordless Authentication.** IJERA 14(7), pp. 58–63, 2024. 10.9790/9622-14075863
- Mitra, A. et al. **/dev/SDB: Software Defined Boot.** arXiv, 2026. *(Accepted & Presented, IEEE CIIT 2023)* 10.48550/arXiv.2601.20629
- Mitra, A. & Sethuraman, S. C. **Verifiable Passkey: The Decentralized Authentication Standard.** arXiv, 2025. *(Accepted & Presented, IEEE CIIT 2023)* 10.48550/ARXIV.2512.21663
- Mitra, A. & Sethuraman, S. C. **The Qey: PQC in FIDO2.** arXiv, 2025. 10.48550/ARXIV.2510.21353
- Sethuraman, S. C., Mitra, A. et al. **Loki: FIDO2 IoT Lock for Physical Asset Security.** IEEE Access, 2022. 10.1109/ACCESS.2022.3216665
- Sethuraman, S. C., Mitra, A. et al. **TUSH-Key: Transferable User Secrets on Hardware Key.** IEEE ICPADS 2024, pp. 176–185. 10.1109/ICPADS63350.2024.00032
- Ghosh, A., Mitra, A. et al. **Saila: HID Injection Protection with Smartphone-based Passwordless Security.** IEEE ICDCSW 2024, pp. 122–127. 10.1109/ICDCSW63686.2024.00023
- Cherukuri, A. K., Shaw, R., Ghosh, A., Mitra, A. et al. **FIDO2 Smart Medical Card for Healthcare.** Int. J. Critical Computer-Based Systems, 11(1/2), 2024. 10.1504/ijccbs.2024.10064420

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